Rates of Convergence - are they optimal?

Sub-gradient method: $x_{t+1} = x_t - \eta g_t, \quad g_t \in \partial f(x_t)$

Gradient descent: $x_{t+1} = x_t - \eta \nabla f(x_t)$

Subgradient method for Lipschitz convex functions

$\hookrightarrow \exists G, \ \|g\|_2 \le G, \ \forall g \in \partial f(x)$

After T iterations

Error : $O(1/\sqrt{T}) \longleftrightarrow$ For error ε need $O(1/\varepsilon^2)$ iterations,

Gradient descent for smooth convex functions

Error : $O(1/T) \longleftrightarrow \varepsilon$ need $O(1/\varepsilon)$

Gradient descent for smooth (β) and strongly (α) convex functions

Error : $O\left(\left(\frac{K-1}{K+1}\right)^T\right) \longleftrightarrow \varepsilon$ need $O(K \log(1/\varepsilon))$

K condition number : (β/α)

Oracle model of computation

What do we know about our function?

$$\min: f(x) \quad \text{st}: x \in \mathcal{X} = \mathbb{R}^d$$

$$x \longrightarrow \boxed{\text{Oracle}} \longrightarrow f(x),\ \nabla f(x),\ g_x \in \partial f(x)$$

Suppose the only access we have to our function f is through the oracle.

Q: How many calls to the oracle do we need in the worst case over a class of functions, in order to guarantee error ε?

Oracle based lower bounds

For the class of Lipschitz convex functions, there is no algorithm which can guarantee error better than $O(1/\sqrt{T})$ for all functions in this class.

$\Rightarrow$ This tells us that the subgradient method is <u>unimprovable</u>.

For smooth functions ..... the lower bound is $O(1/T^2)$

For β-smooth and α-strongly convex functions ... the lower bound is:
$$O\left(\left(\frac{\sqrt{K}-1}{\sqrt{K}+1}\right)^T\right)$$

$K = \beta/\alpha \geq 1 \Rightarrow \sqrt{K} < K$

can we find an algorithm to match.

Discussion about the oracle model

Is the assumption reasonable? What is the takeaway?

$x \longrightarrow \boxed{\text{Oracle}} \longrightarrow f(x), \nabla f(x)$

If we develop any algorithm that only uses evaluations of $f(x)$ and $\nabla f(x)$, then we are bound by the oracle-lower-bounds.

Ex: $\boxed{\begin{matrix} x_1, \dots, x_{t-1}, x_t, x_{t+1} \\ \nabla f(x_1) \dots\dots \end{matrix}} \longrightarrow x_{t+2}.$